

Mapping collective flow with femtoscscopy



IRL NPA Workshop, 8-12 December 2025

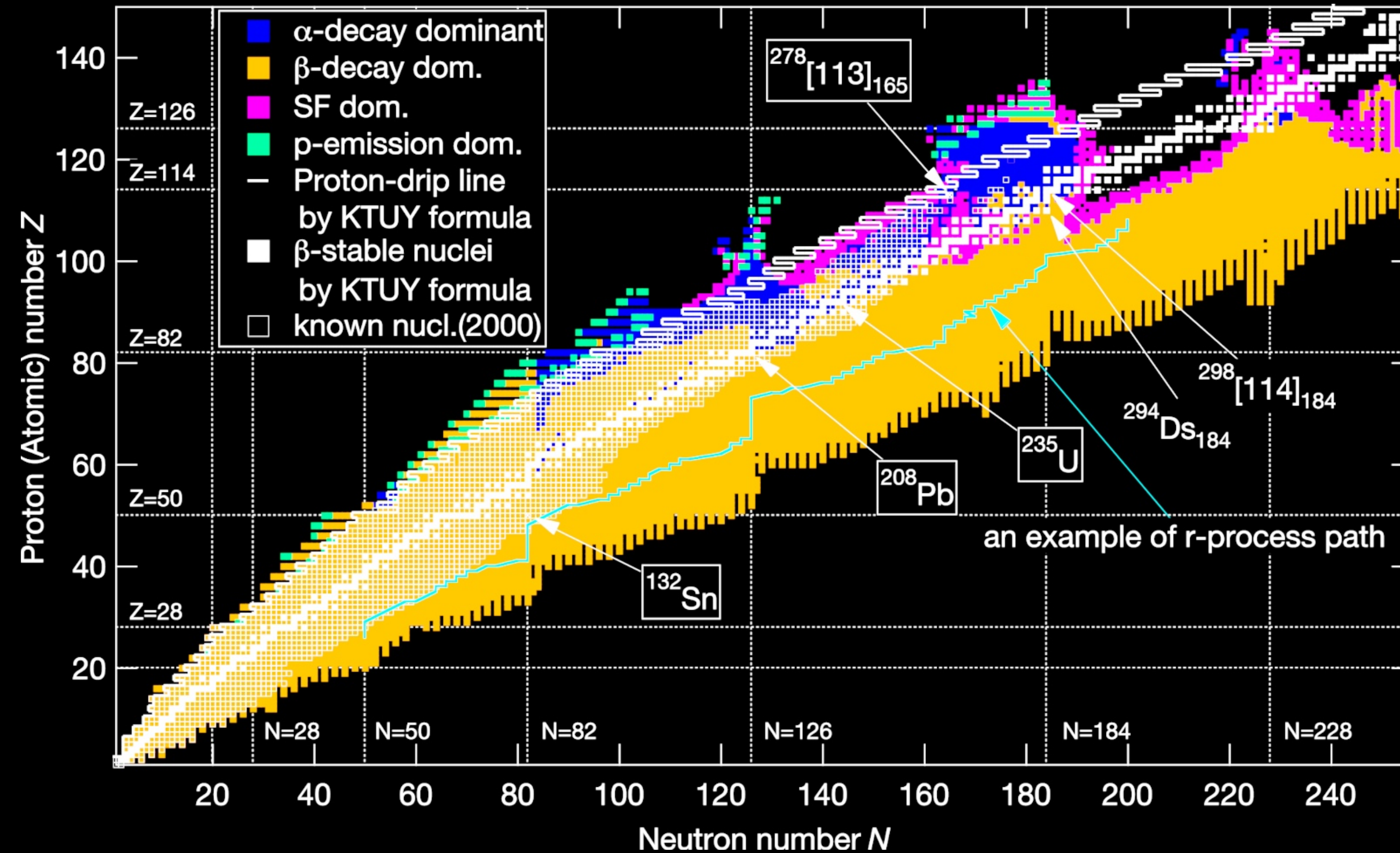
Oleh Savchuk



Facility for
Rare Isotope Beams
at Michigan State University

(Dense) nuclear matter

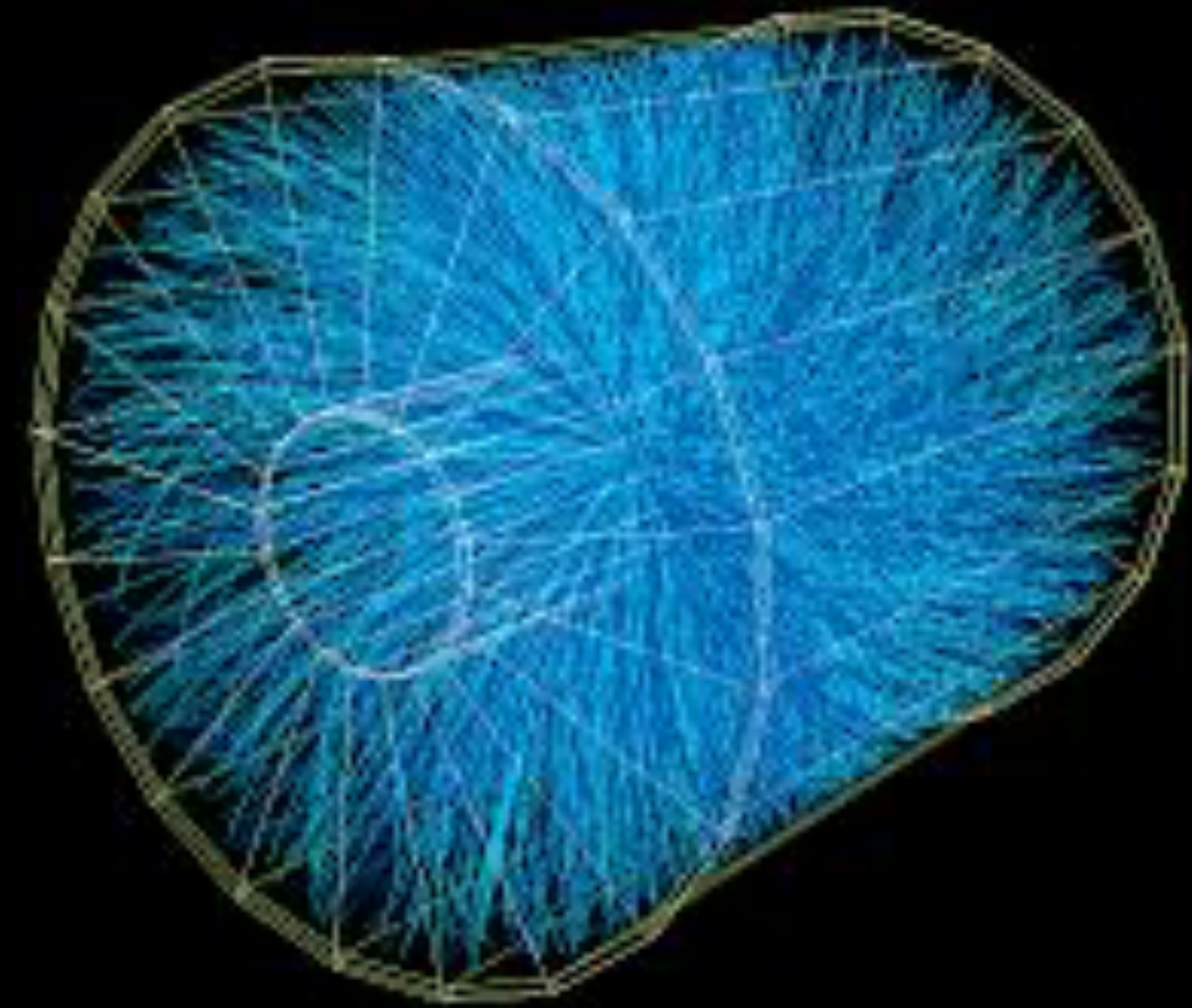
In the universe



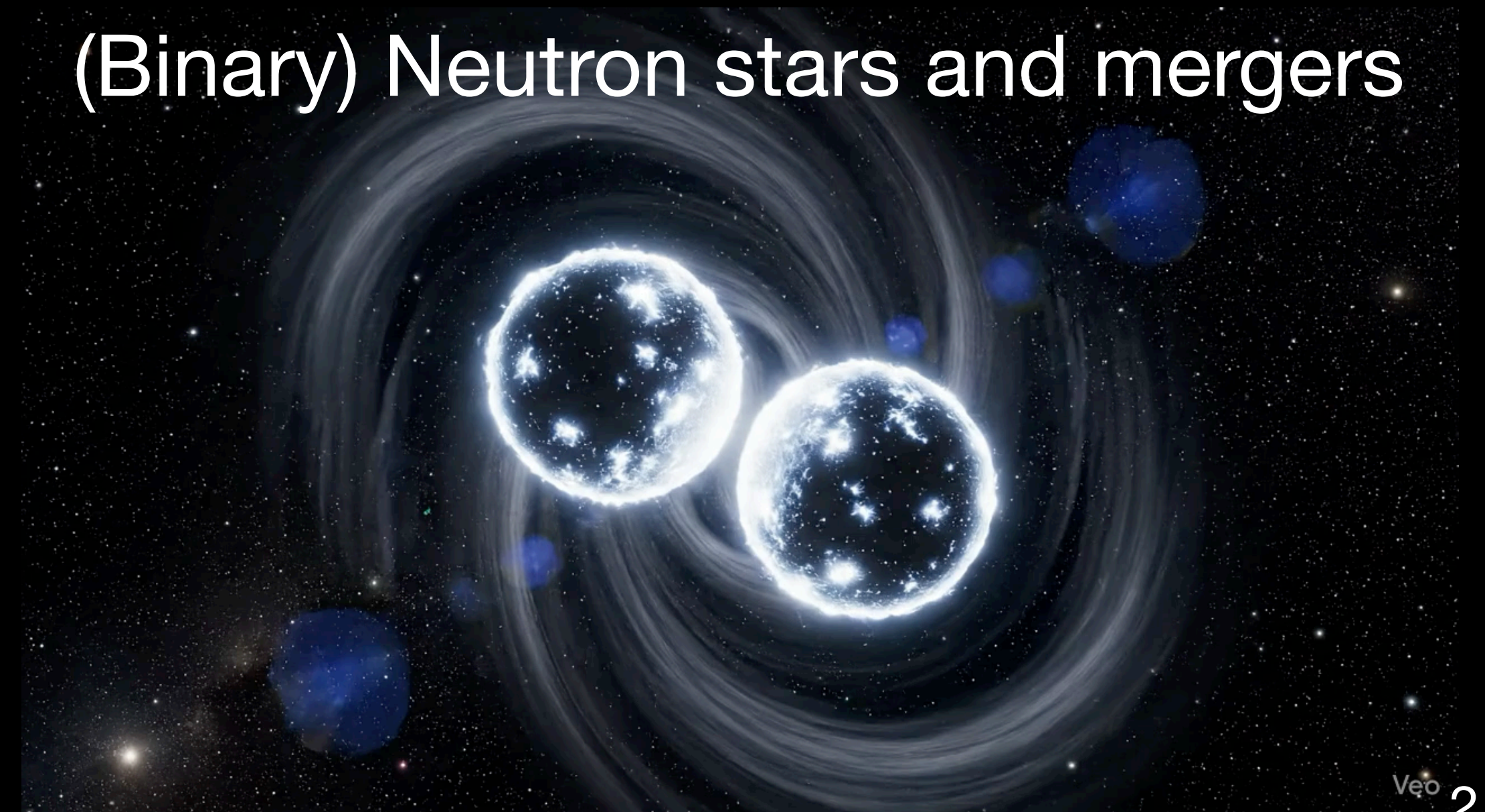
Building block of the universe.

Poorly understood under “extreme” conditions

Heavy-Ion Collisions

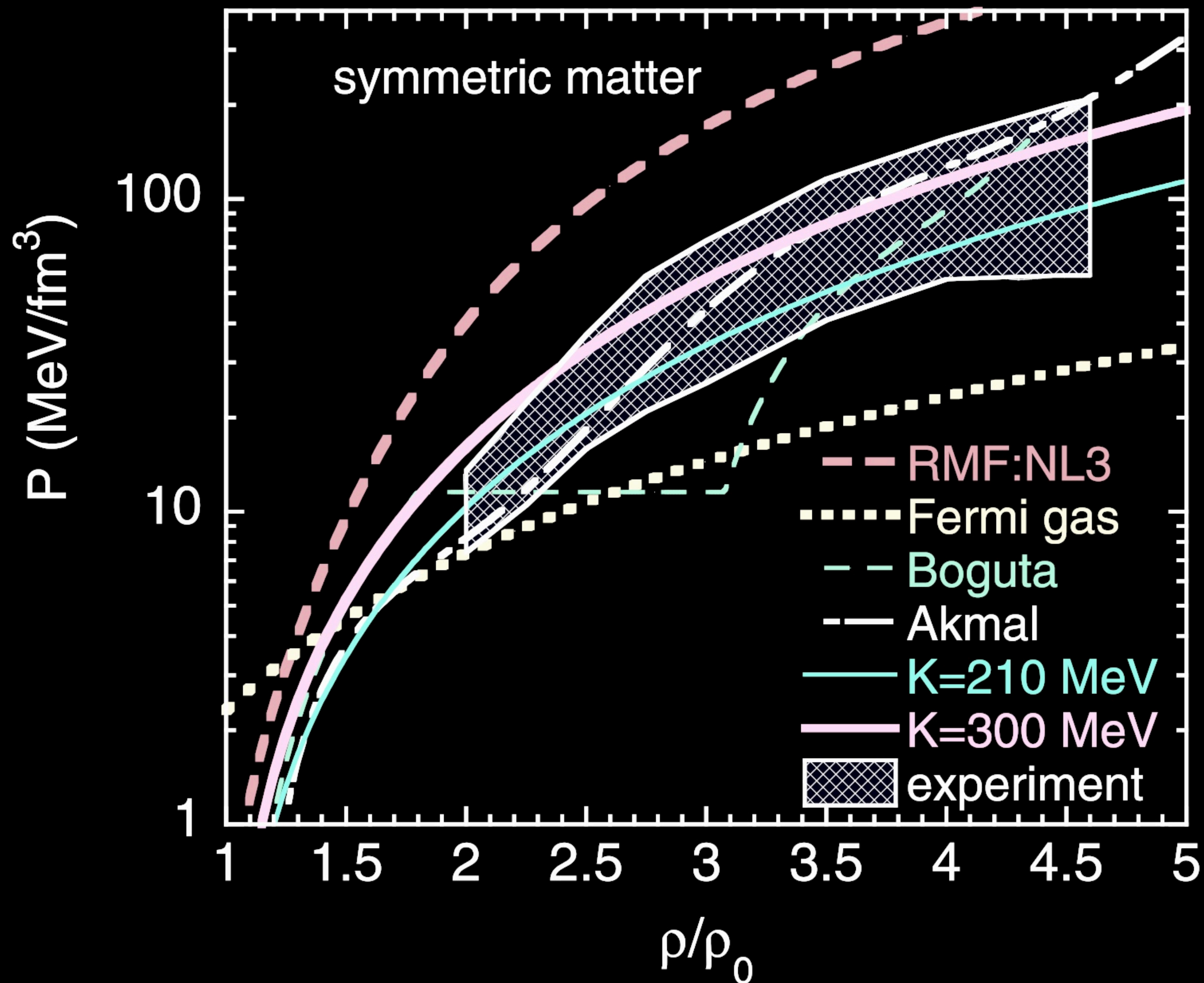


(Binary) Neutron stars and mergers



Dense Nuclear Matter

Equation of state



EoS contains most basic information about matter's properties

EoS is poorly understood beyond nuclear saturation $\rho_0 \approx 0.16 \text{ fm}^{-3}$

But highly important for Neutron Star including gravitational wave signals

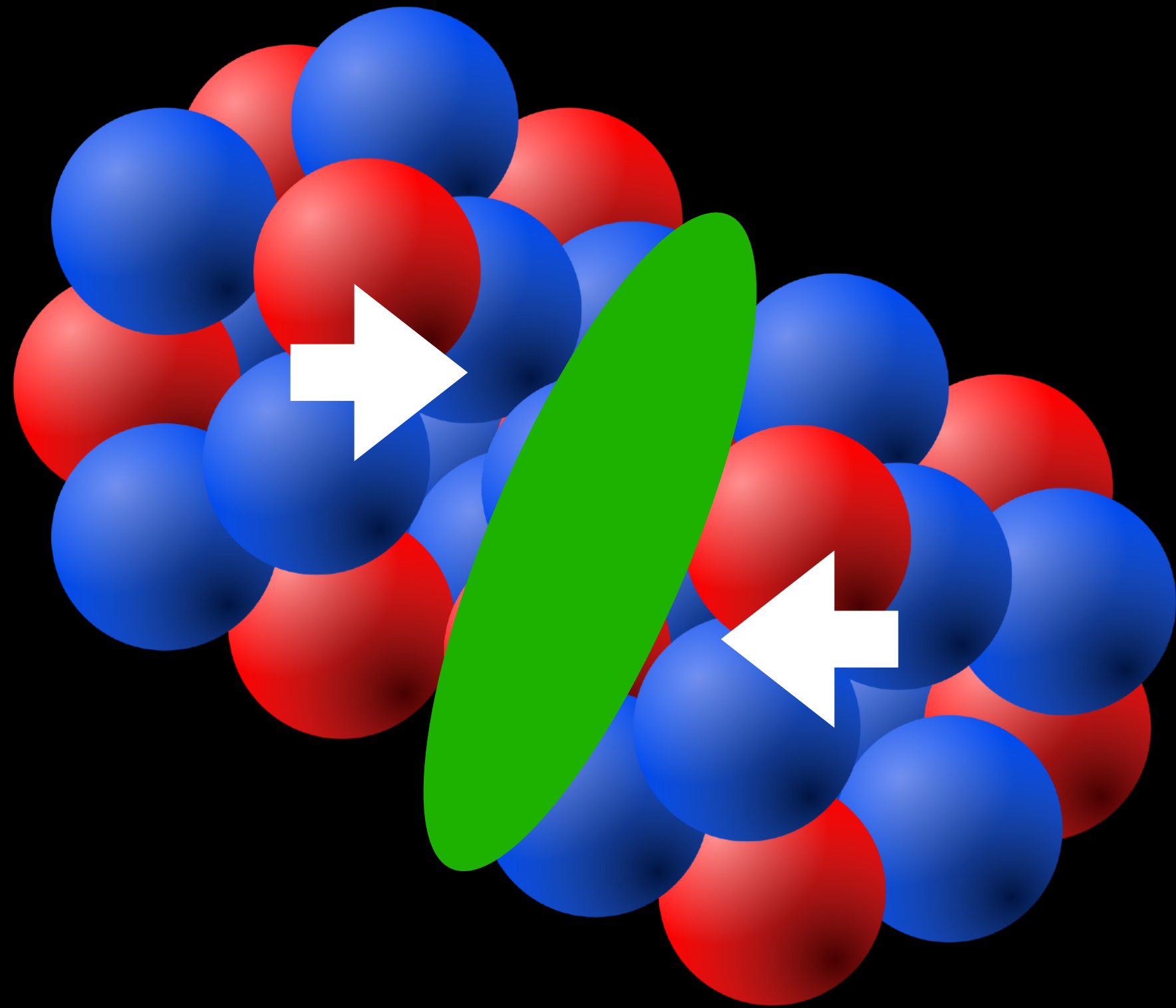
It was shown by Danielewicz, Lacey, Lynch that it is possible to extract EoS from HIC

[Danielewicz, Lacey, Lynch, Science 298 (2002) 1592-1596]

Dense Nuclear Matter

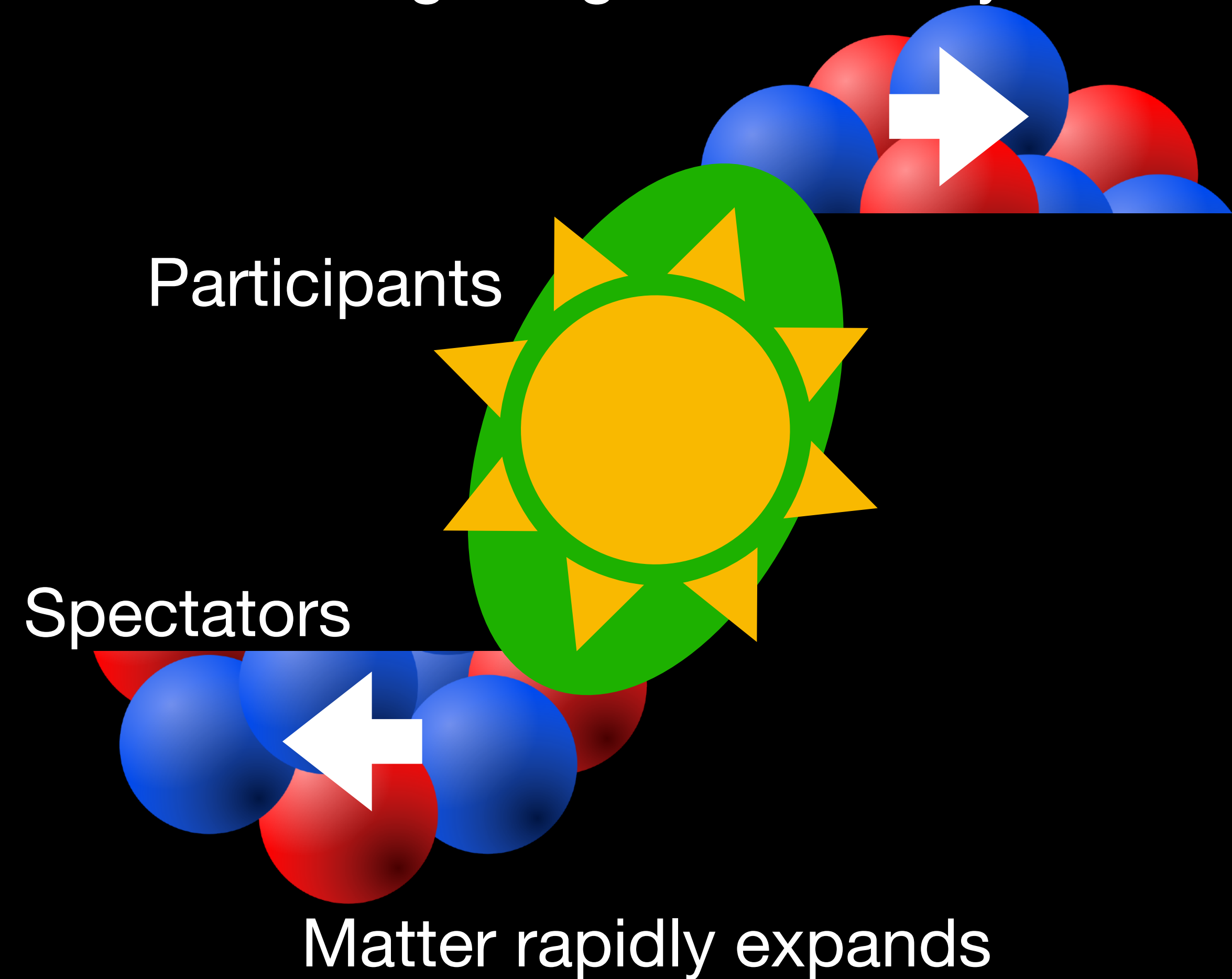
Standart model of Heavy-Ion Collision

Deposition of energy into overlap region

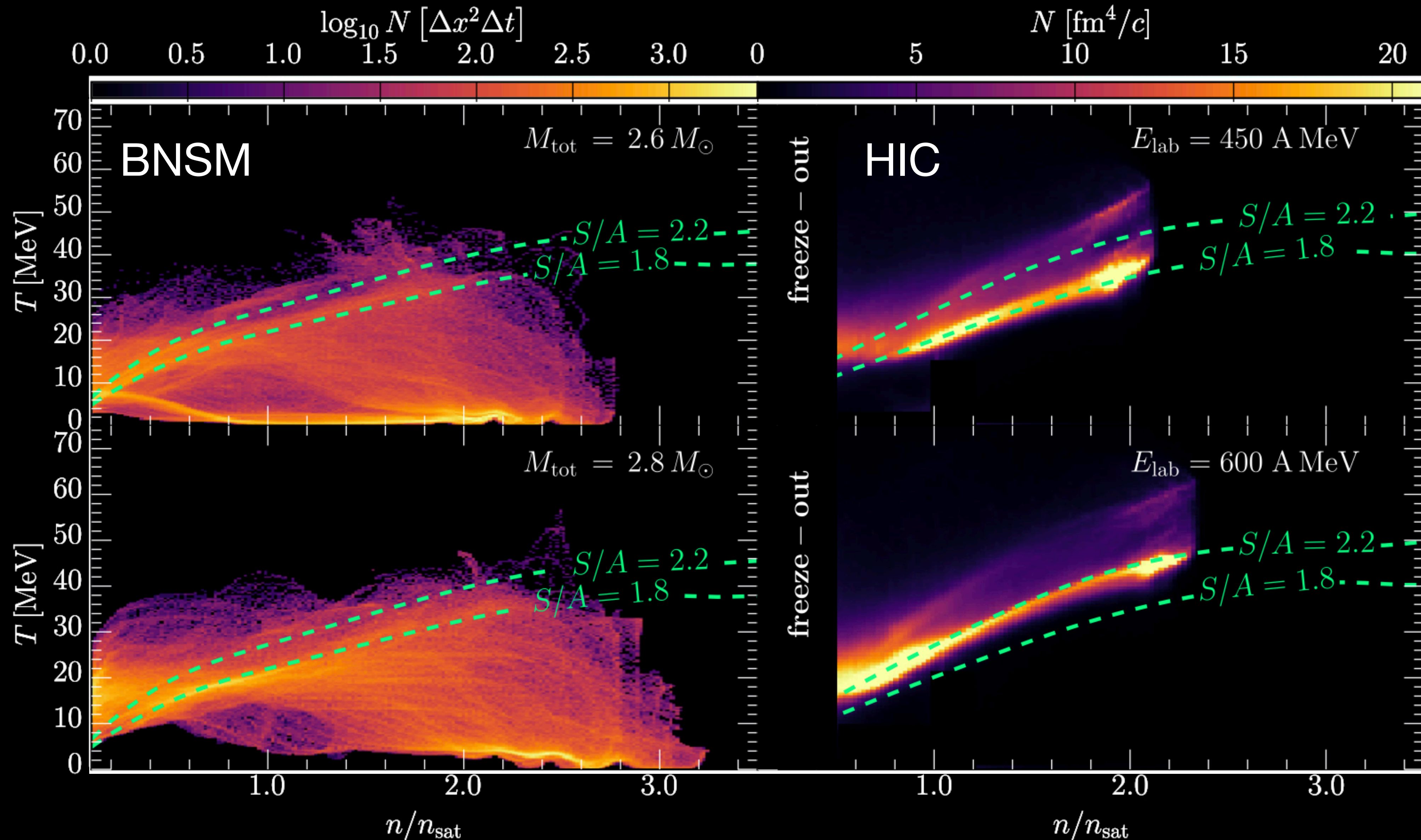


Creates high temperature and density

Creates tiny explosion
“Big bang in laboratory”



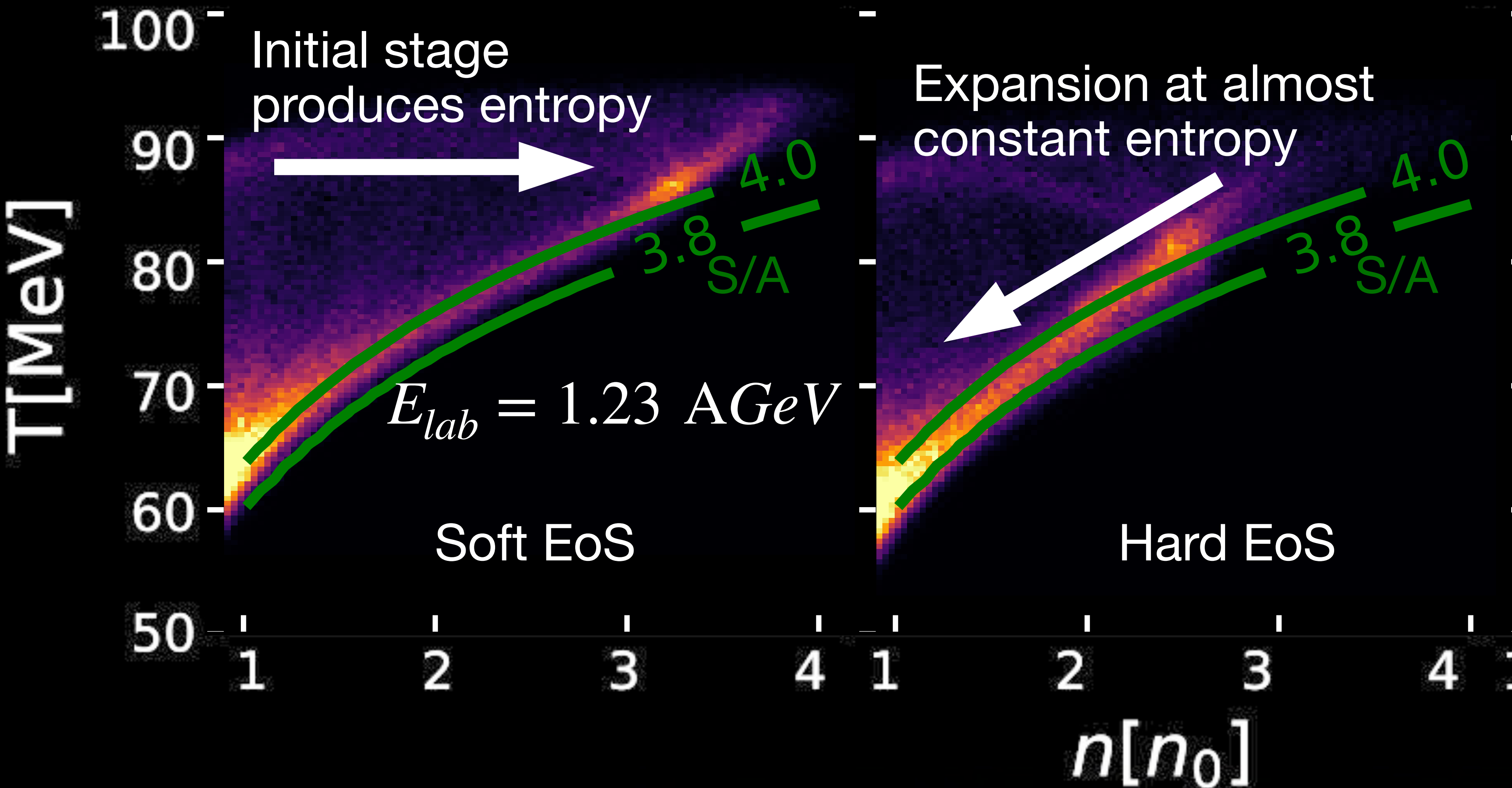
Neutron stars vs Heavy Ion Collisions



Hydro models show that HIC probe entropy similar to the one expected from BNSM

At densities reached at future facilities

Coarse Graining Transport Simulations



Coarse graining transport simulations confirms hydrodynamics

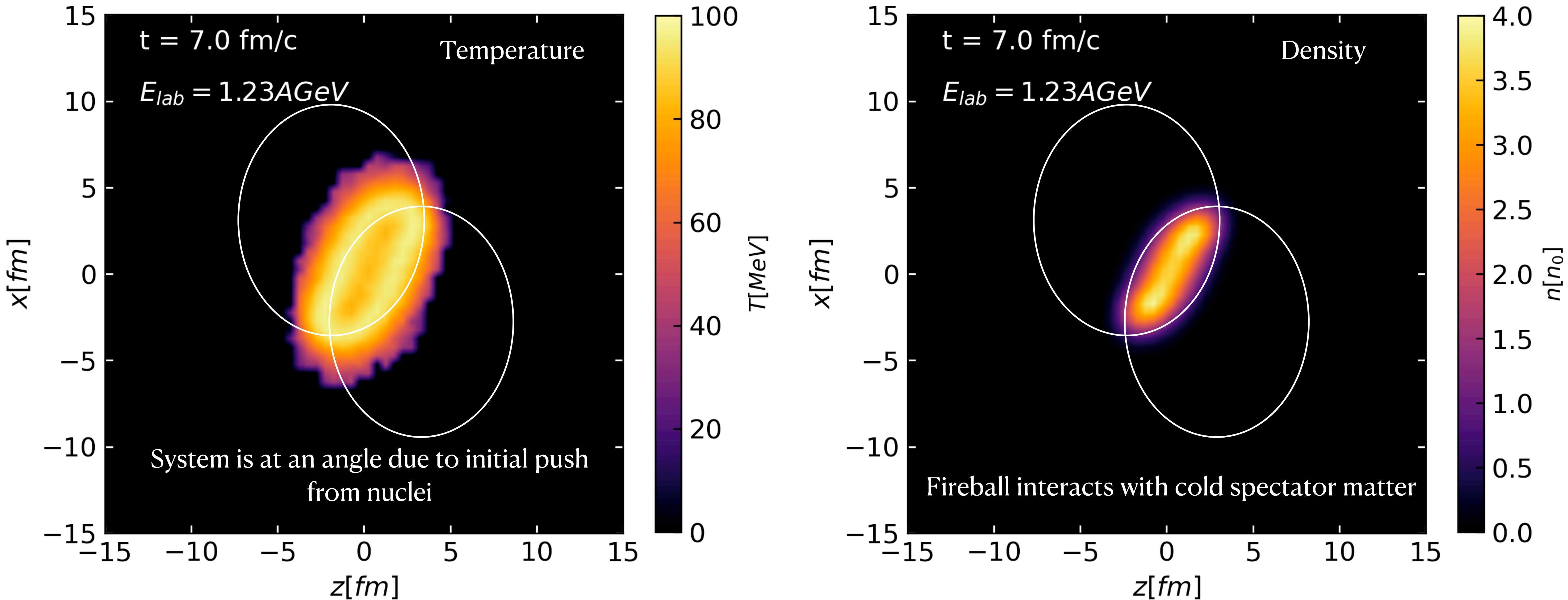
Advantage - non-equilibrium effects included together with microscopic interactions

HADES and FRIB probe $2 - 3n_0$, with FRIB allowing to probe symmetry energy dependence

Phenomena in mid-central collisions

UrQMD, CMF-potential, AuAu at $b=6.6$ fm, 15-25% centrality bin

ZX plane (looking from the side)

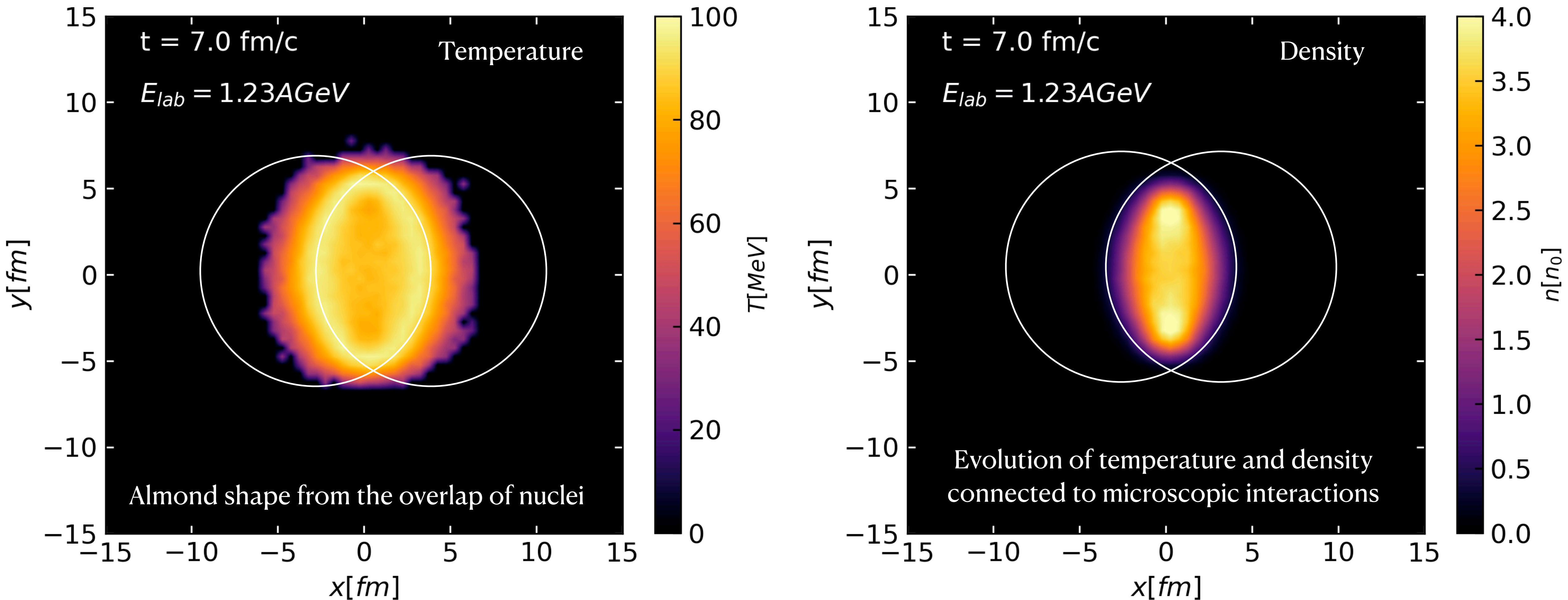


Details like the push in longitudinal direction, presence of cold spectator matter are important!

Phenomena in mid-central collisions

UrQMD, CMF-potential, AuAu at $b=6.6$ fm, 15-25% centrality bin

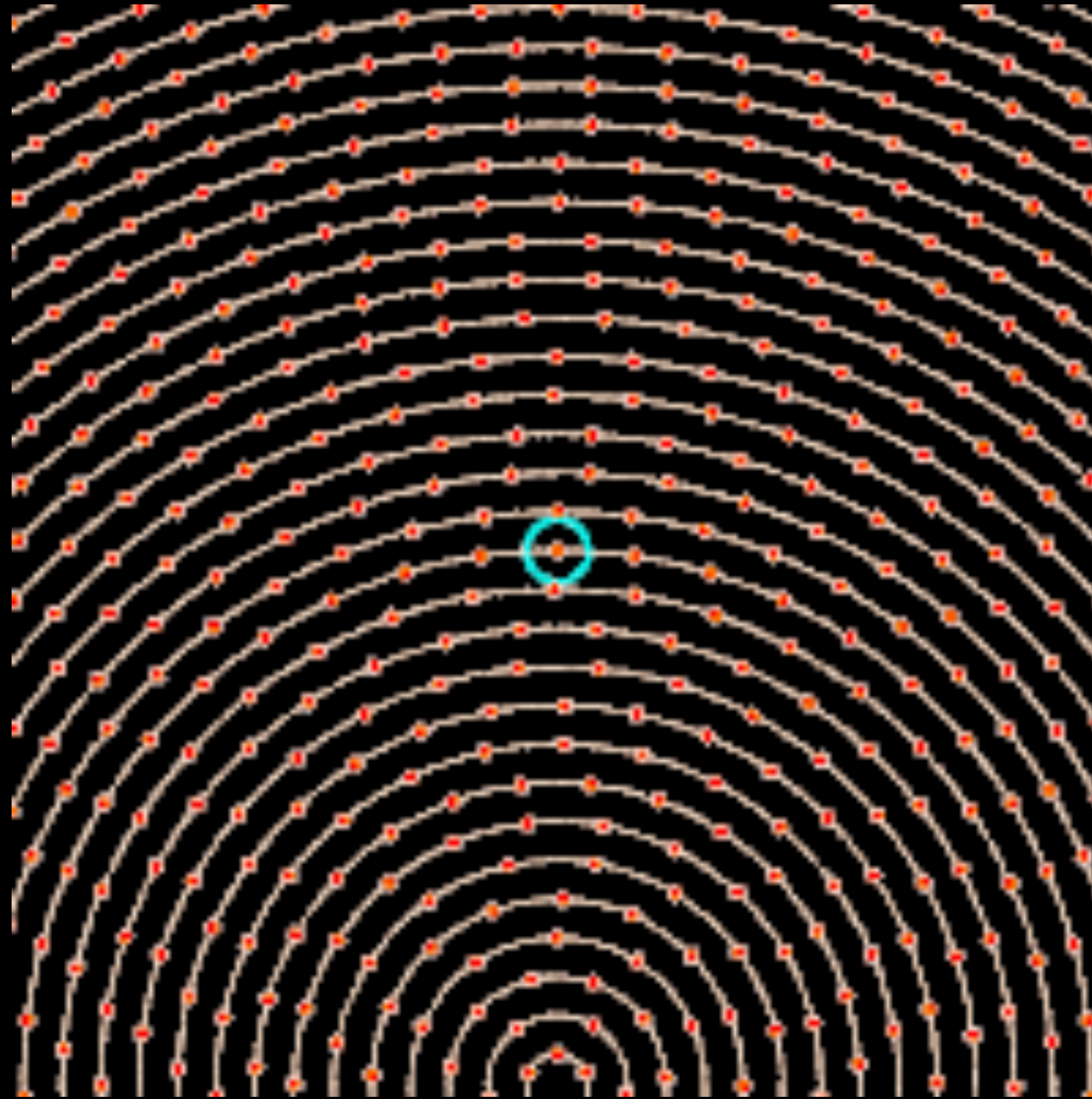
Transverse plane (looking down the beam pipe)



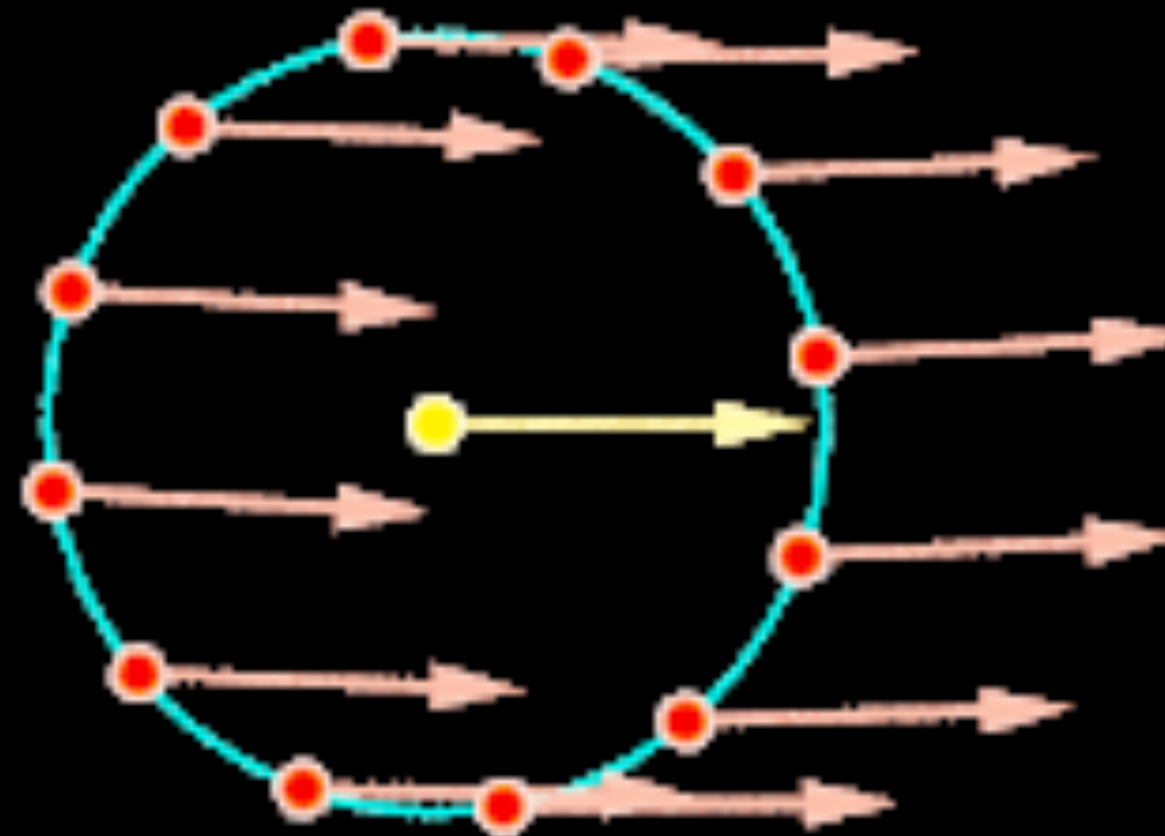
Pressure transforms initial geometry into momentum spectra of particles detected after collision.

Vorticity

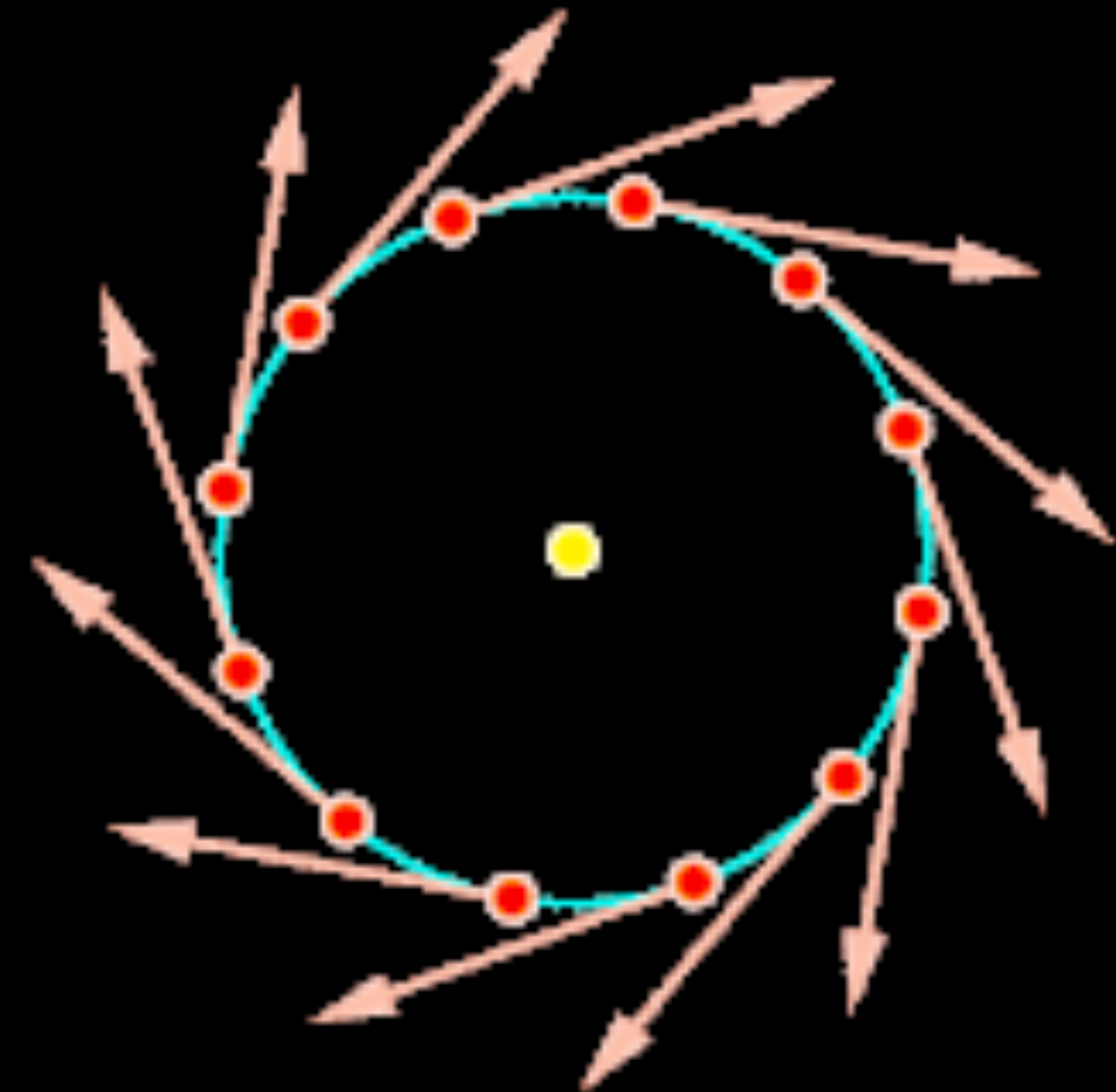
- Vorticity: $\vec{\omega} = [\nabla \times \vec{v}]$ corresponds to local rotation in a fluid.



$$v_{\perp} \sim r$$



Velocity



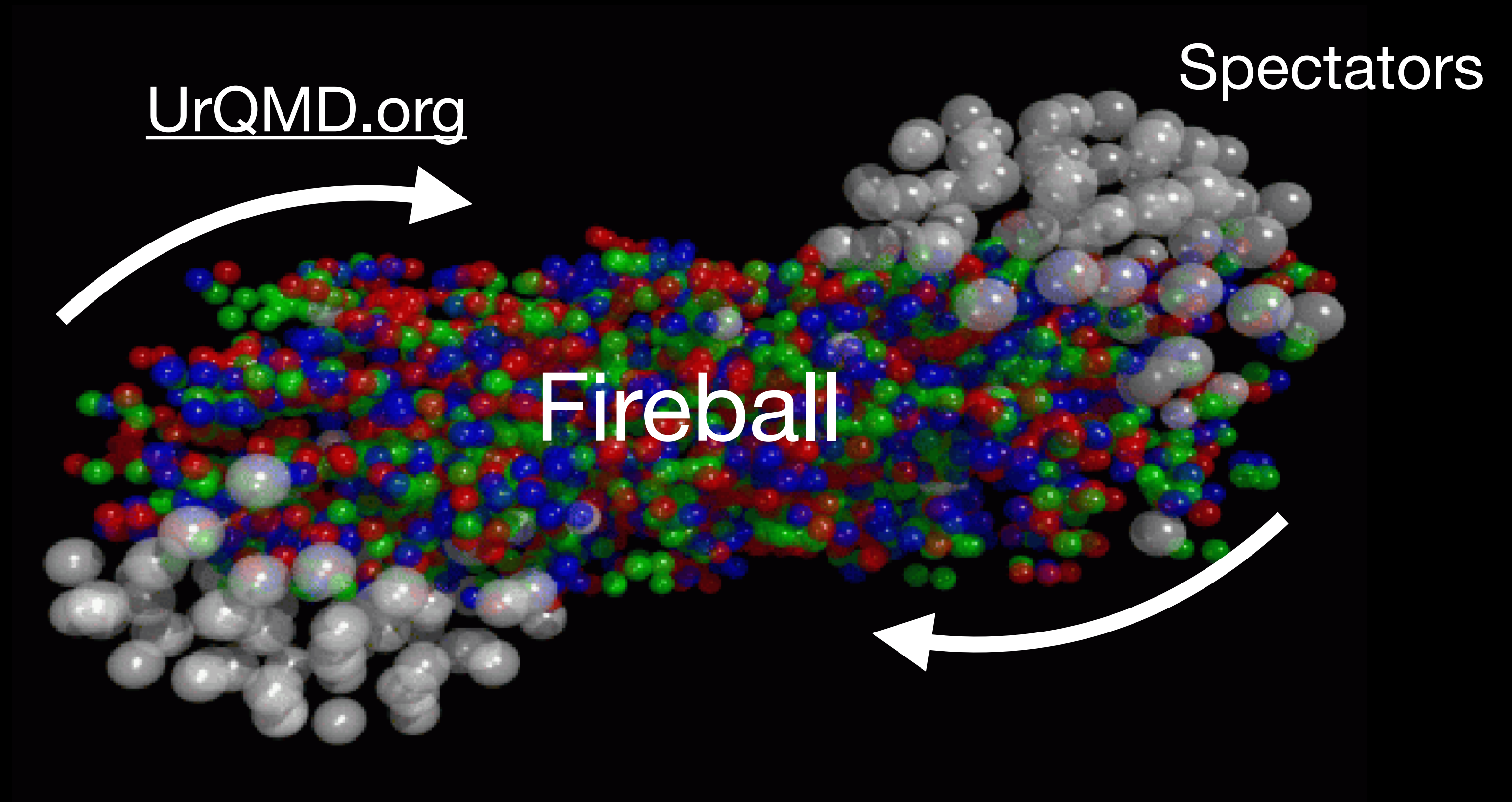
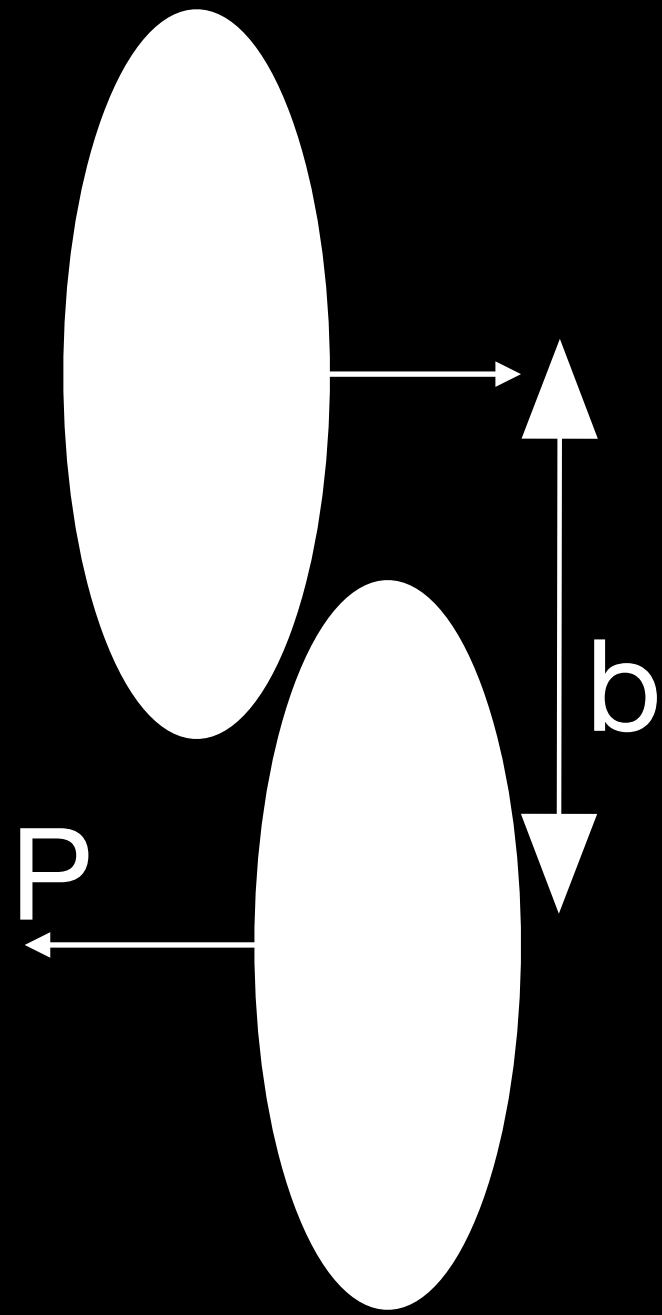
Relative velocity

[By Jorge Stolfi, CC BY-SA 3.0]

- Vorticity is approximately conserved, closely connected with angular momentum.
- Heavy-ion collisions generate one of the most vortical fluids.

Vorticity in Heavy-Ion Collisions

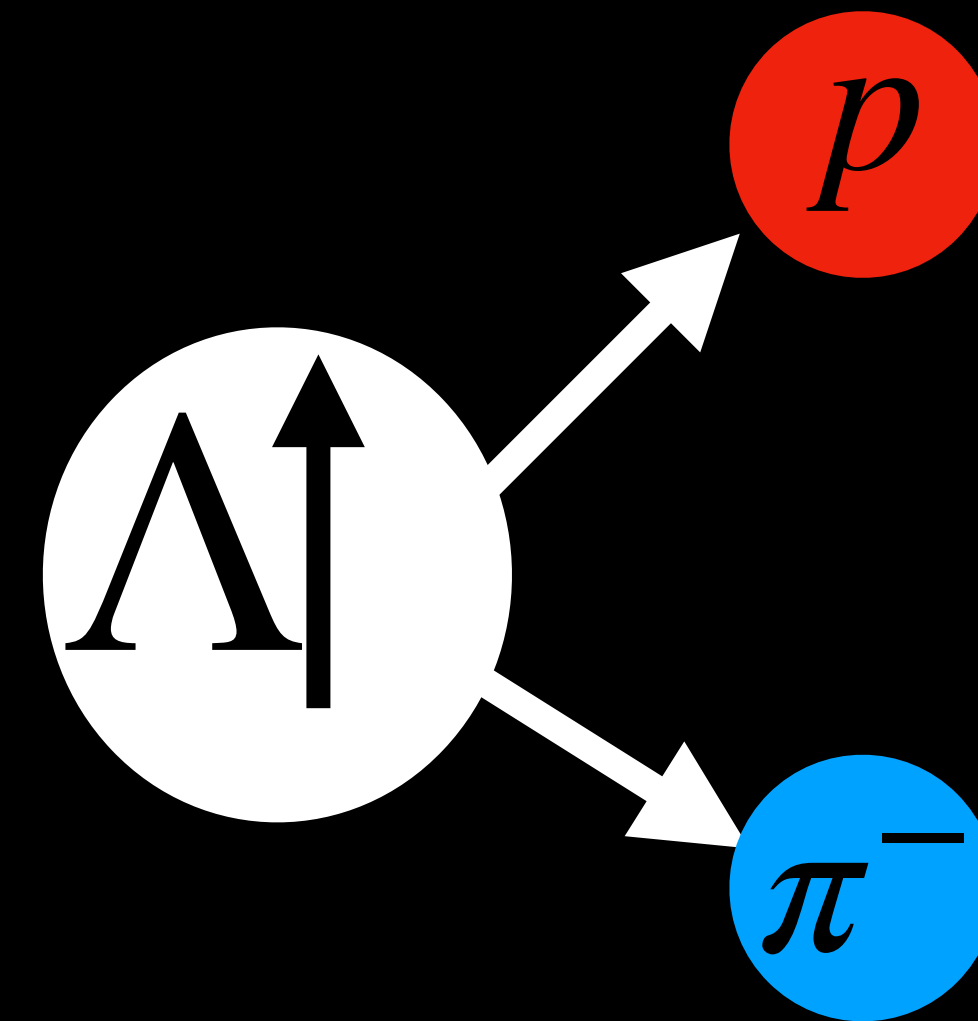
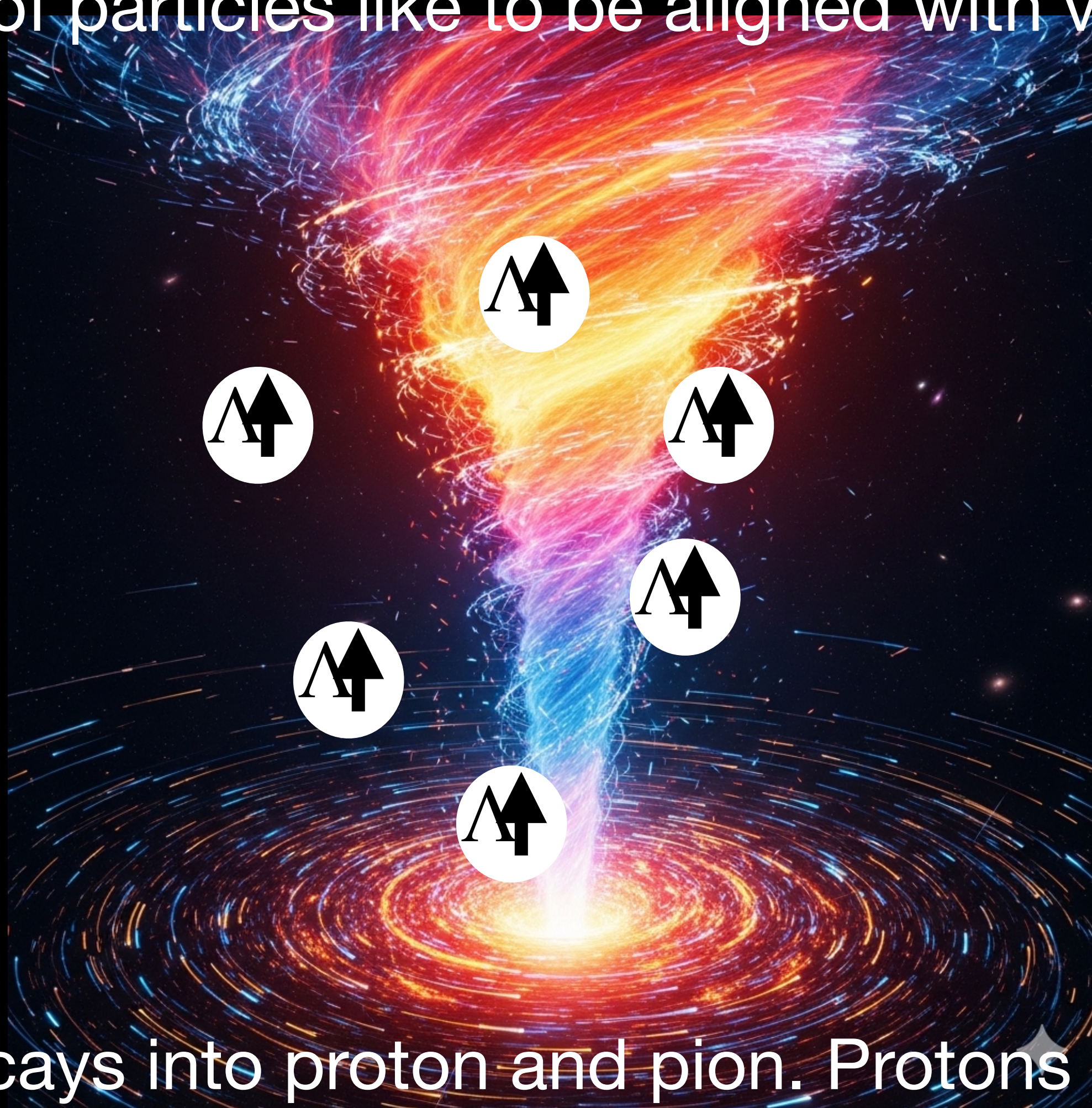
- Non-central collisions have large angular momentum: $L_y = -|bp|$



- Angular momentum grows with energy.
- Vorticity too, but at higher energies most of it is located near spectator.

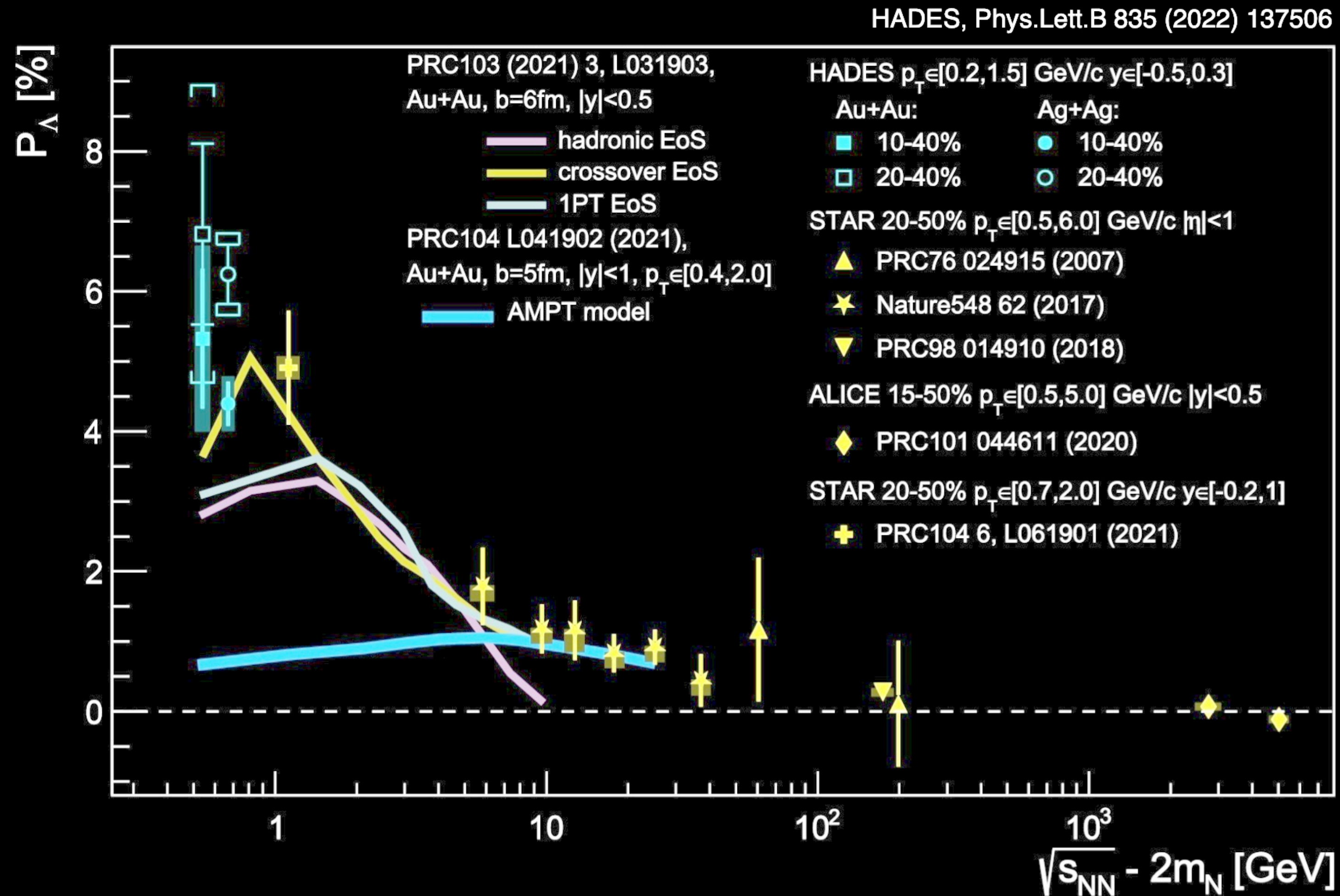
Λ polarization

- Spins of particles like to be aligned with vorticity - polarization in medium.



- Λ decays into proton and pion. Protons more likely to fly in the spin direction.
- Polarization is assumed to be in equilibrium, but medium is dynamic.
- Can we avoid spin relaxation physics?

Λ polarization in Heavy-Ion Collisions



- Low energies are much more affected by vorticity at mid-rapidity
- How to measure below Λ threshold?

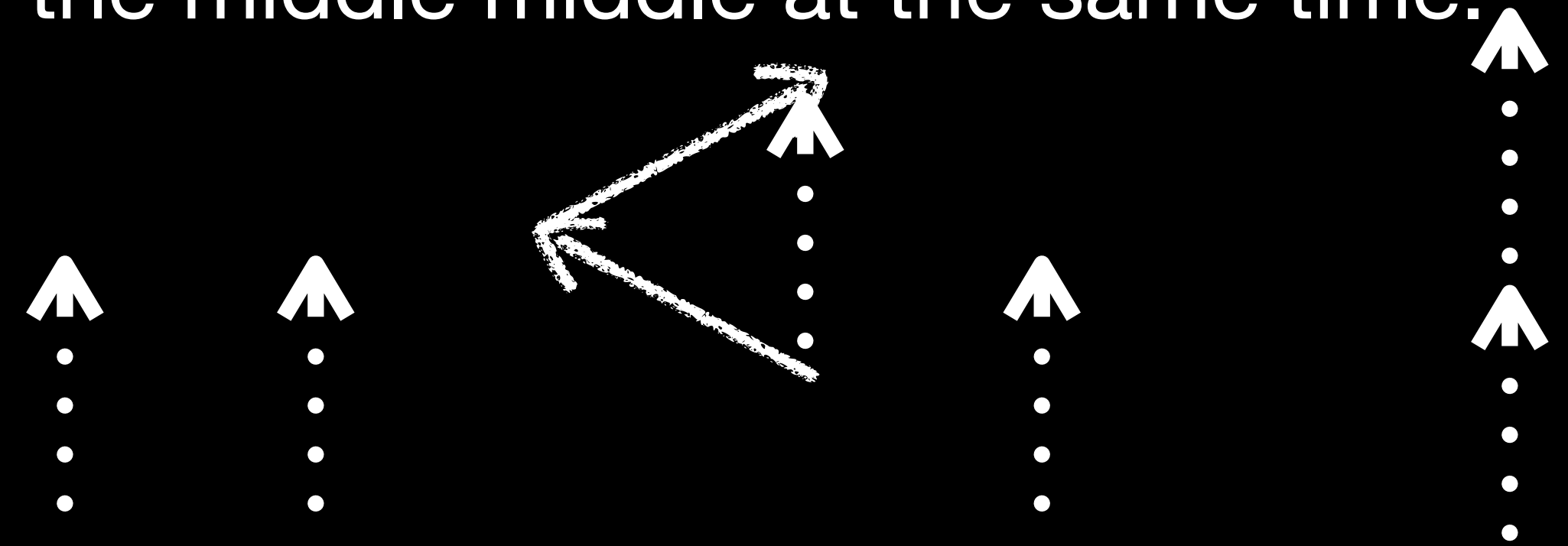


Emission patterns

- If you let go of something on a carousel it's just going straight, tangent to the circle.
- If you throw something it will go between the direction of your throw and circles tangent.



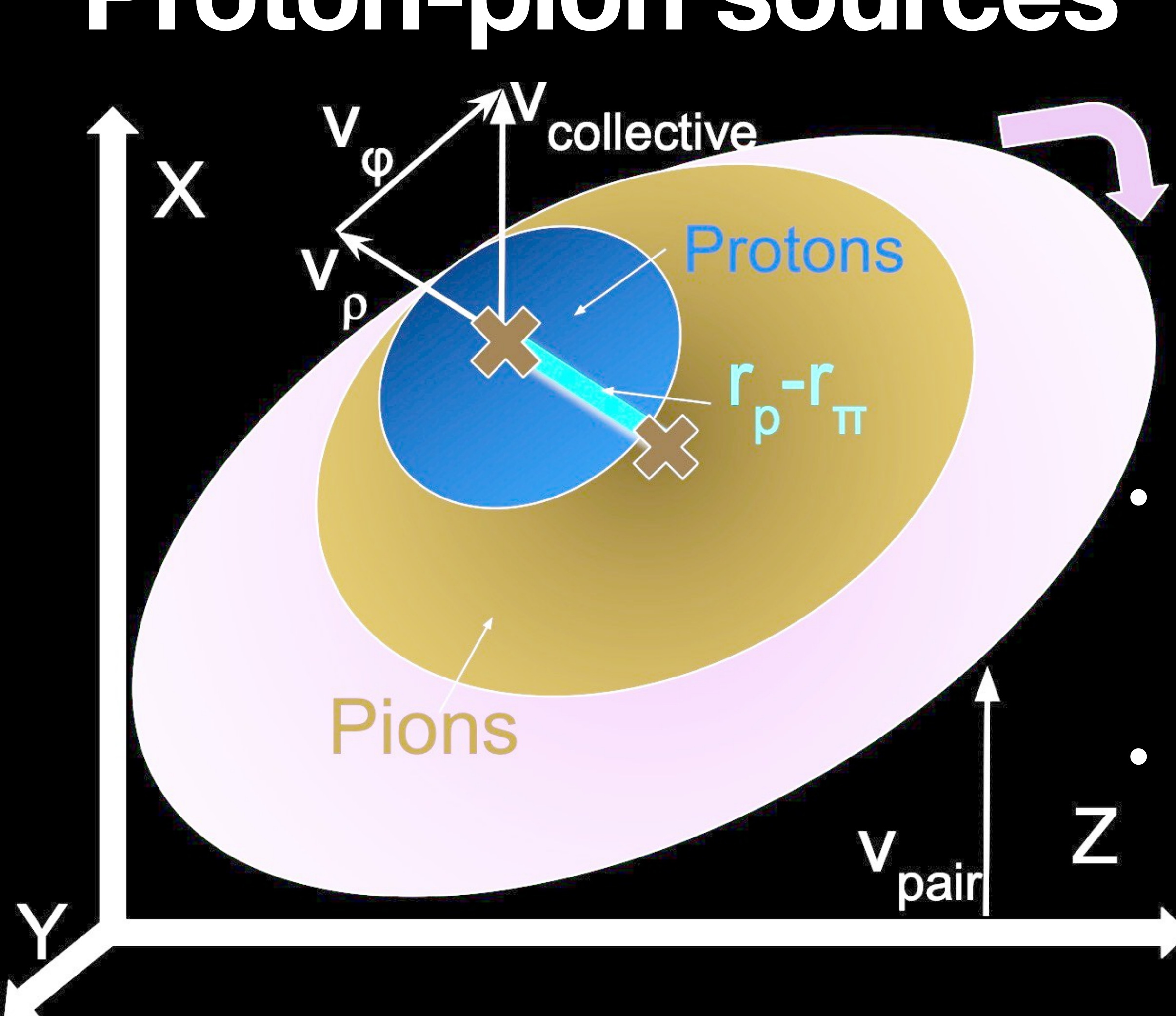
- Ask your friend to throw something from the middle middle at the same time.



Just rotating + throwing out Just throwing

- The resulting distance between projectiles sensitive to the radial and tangential velocities.
- In HIC light particles are emitted from everywhere while heavier have momentum-coordinate correlation - similar to this example.

Proton-pion sources



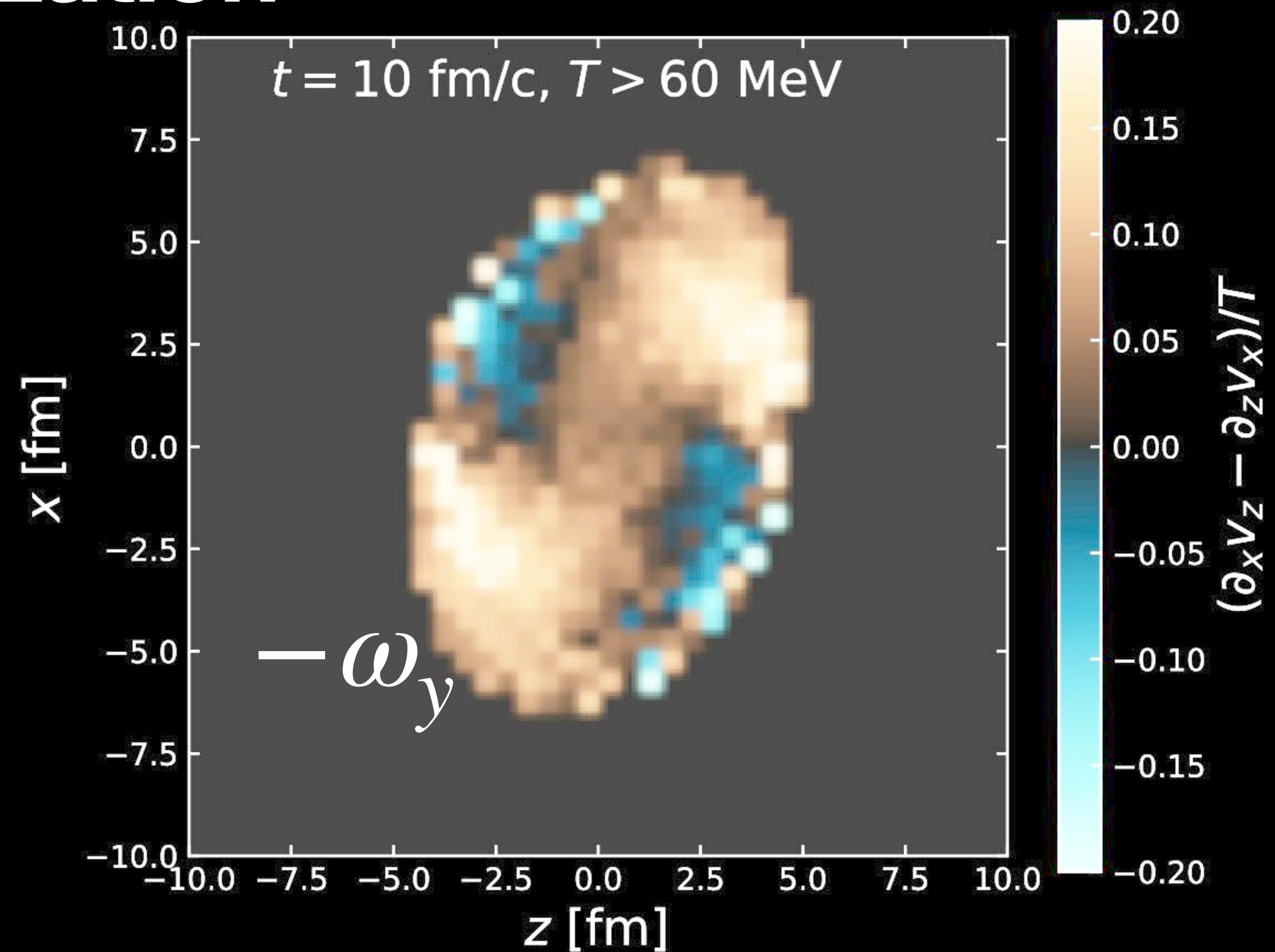
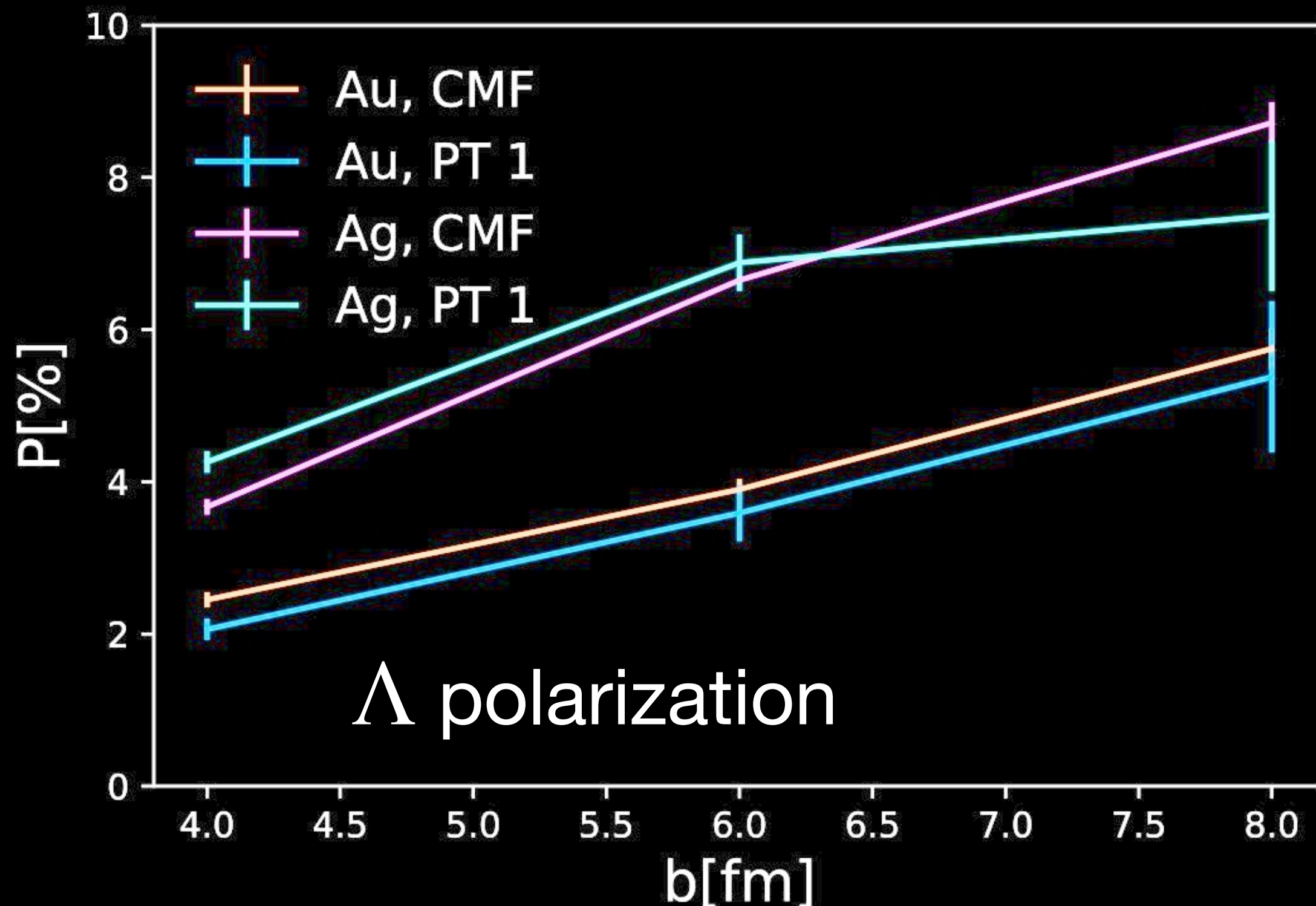
Just rotating + throwing out Just throwing

- Distance between protons and pions at emission is correlated with rotational movement of the fireball.
- In particular **non-zero angle between ejecting velocity and separation within the pair - clear signature of vorticity.**

- Transport simulations are required to understand whether this scenario is feasible in “real” heavy-ion collision.
- To that extend Ultrarelativistic Quantum Molecular Dynamics is used.

Simulations Λ polarization

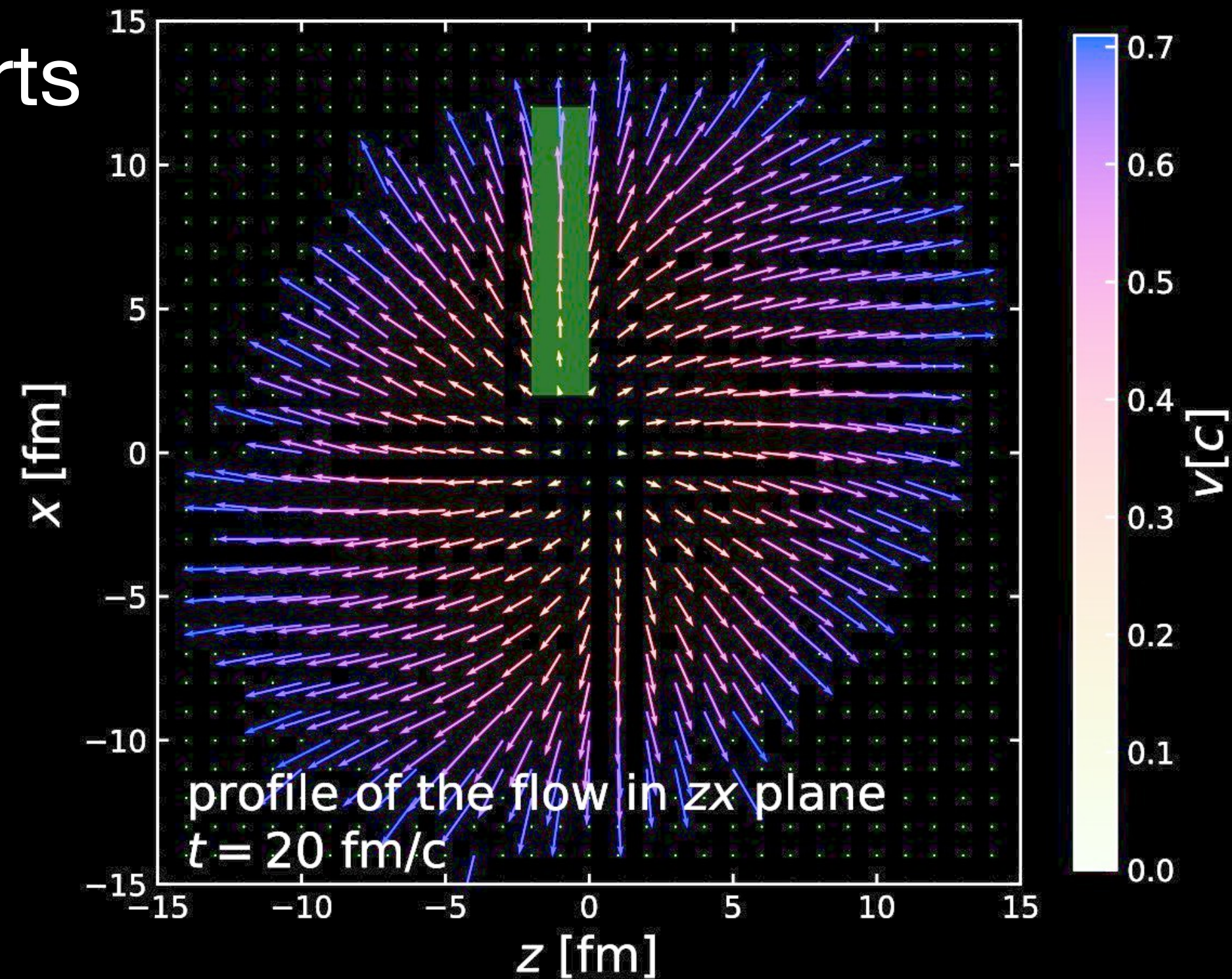
- Overall negative vorticity of the fireball
- Largest near spectators
- Observed thermal lambda polarization with mild sensitivity to the equation of state



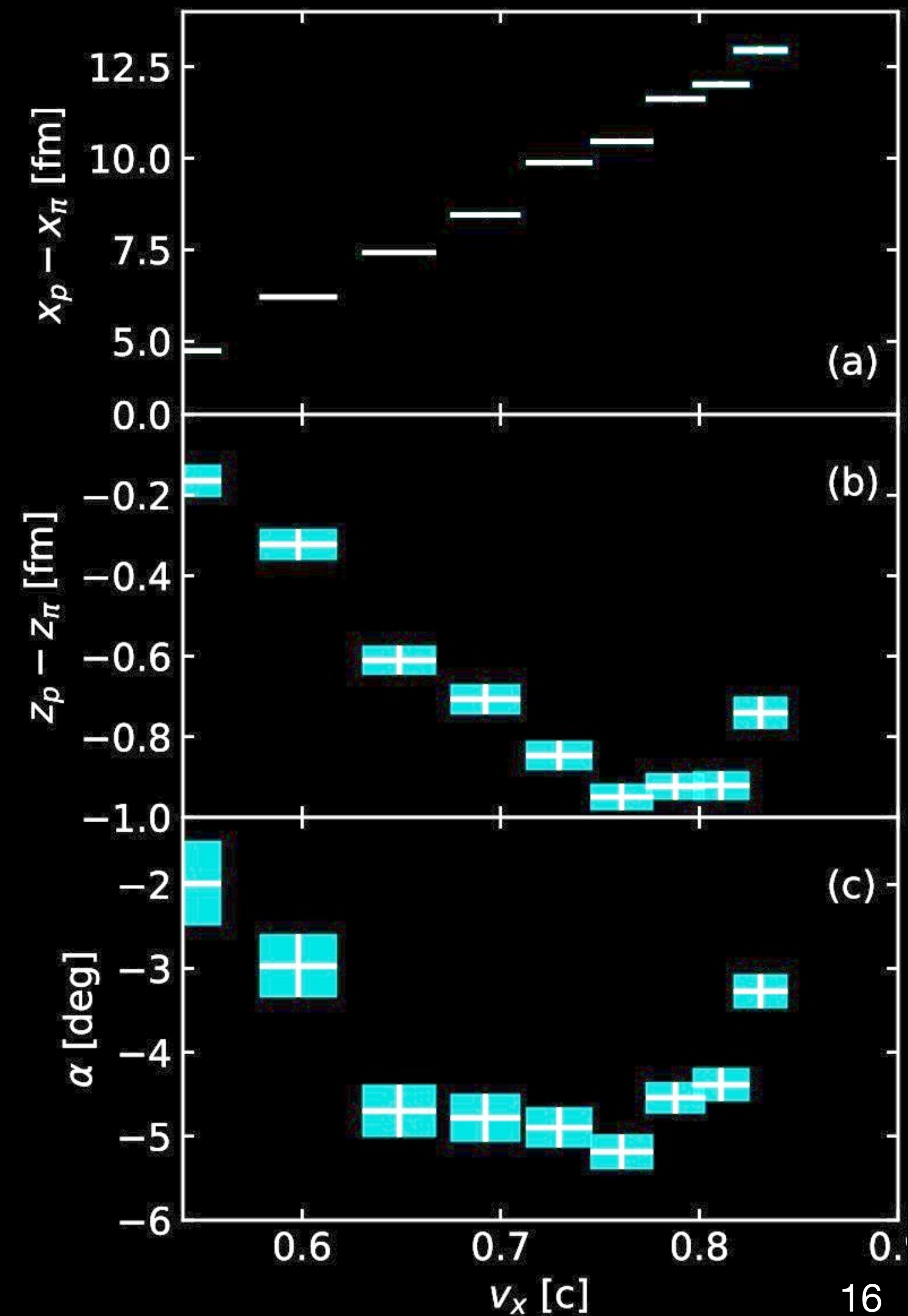
Vorticity evolves with time
Strongest early on, weakens over time

Simulations of emission

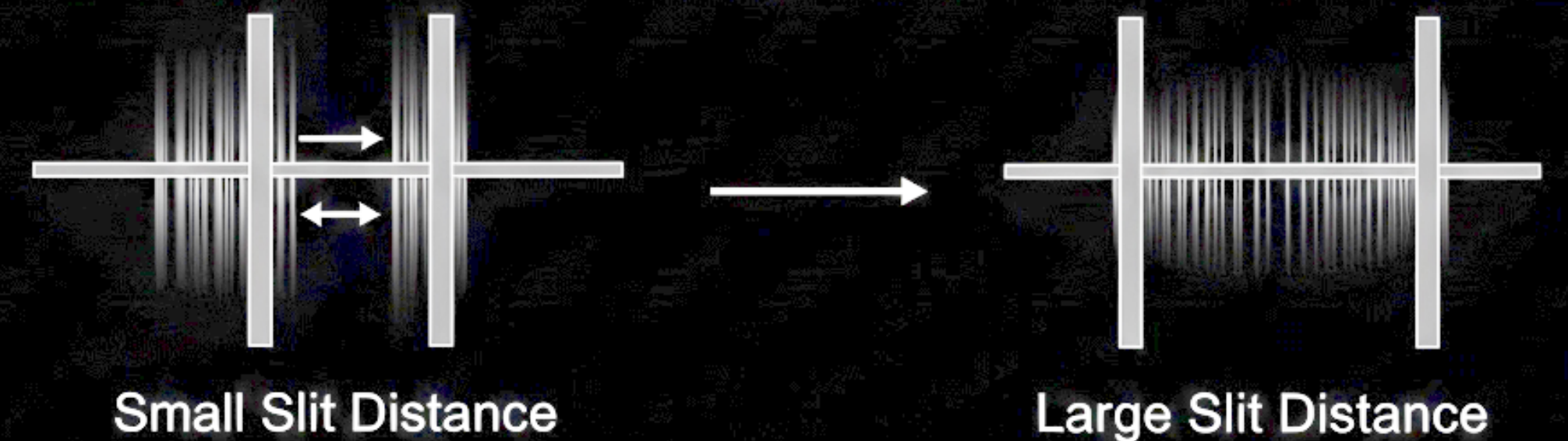
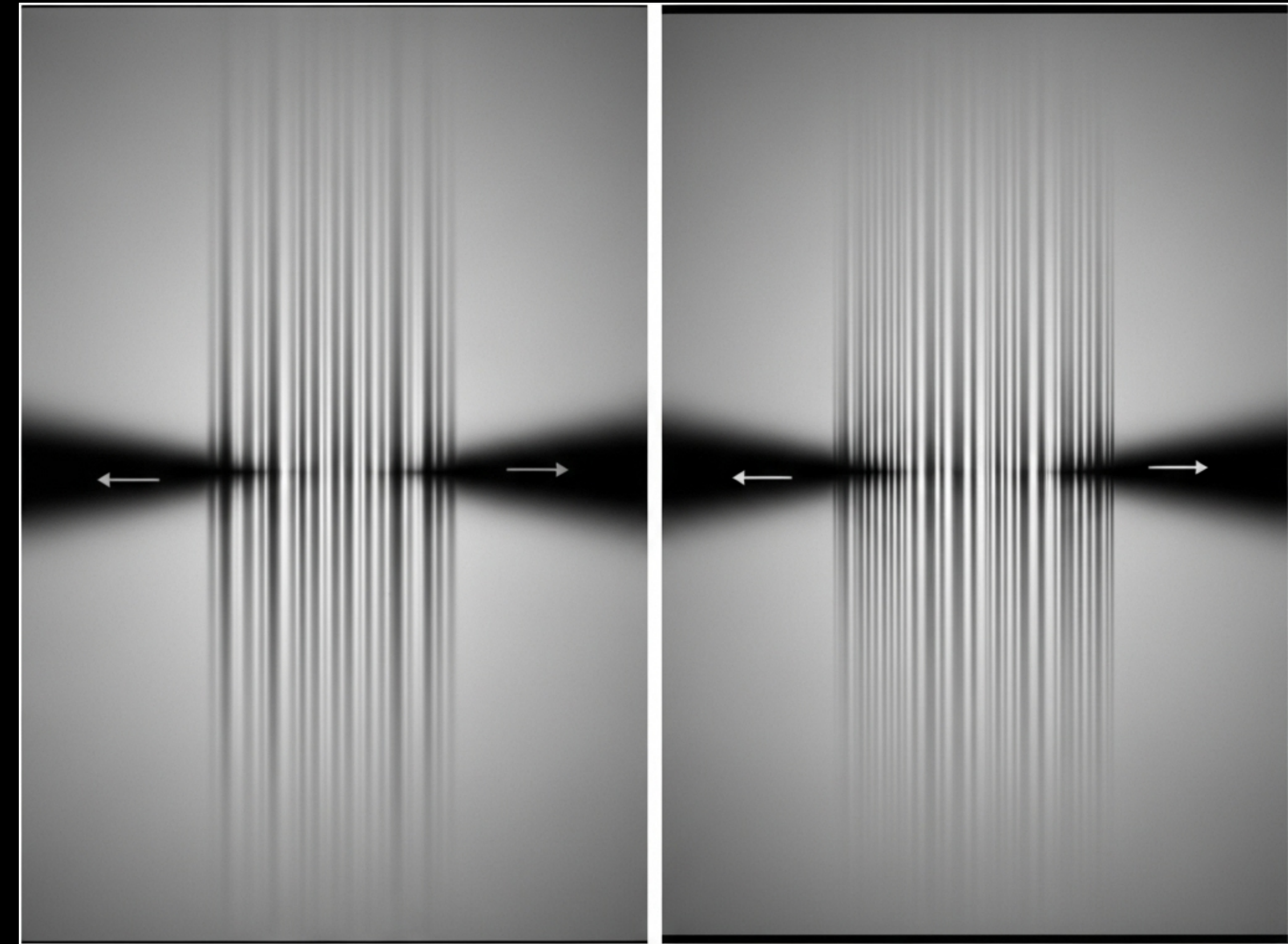
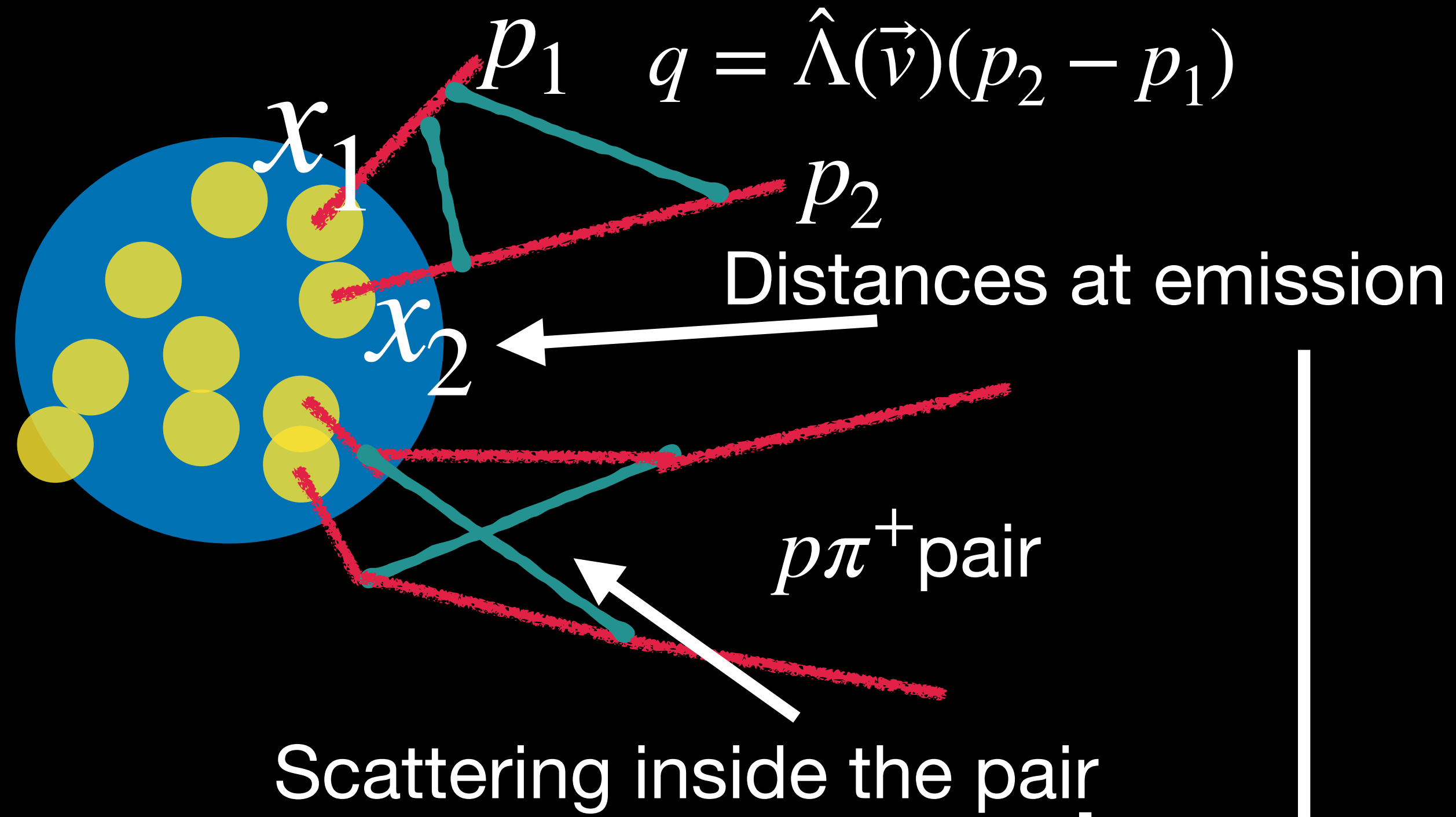
- Flow profile supports that particles flying up emitted at the finite angle to their ejection velocity.



- This angle was calculated and is on the order of 5 degrees.
- Experimentally only finite state momentum of particles is being measured. Therefore one needs to use advanced techniques such as femtoscopy in order to extract it.



Femtoscopy



Koonin-Pratt formula: $C_{\vec{v}}(\vec{q}) = \int d\vec{r} K(\vec{q}, \vec{r}) S_{\vec{v}}(\vec{r})$

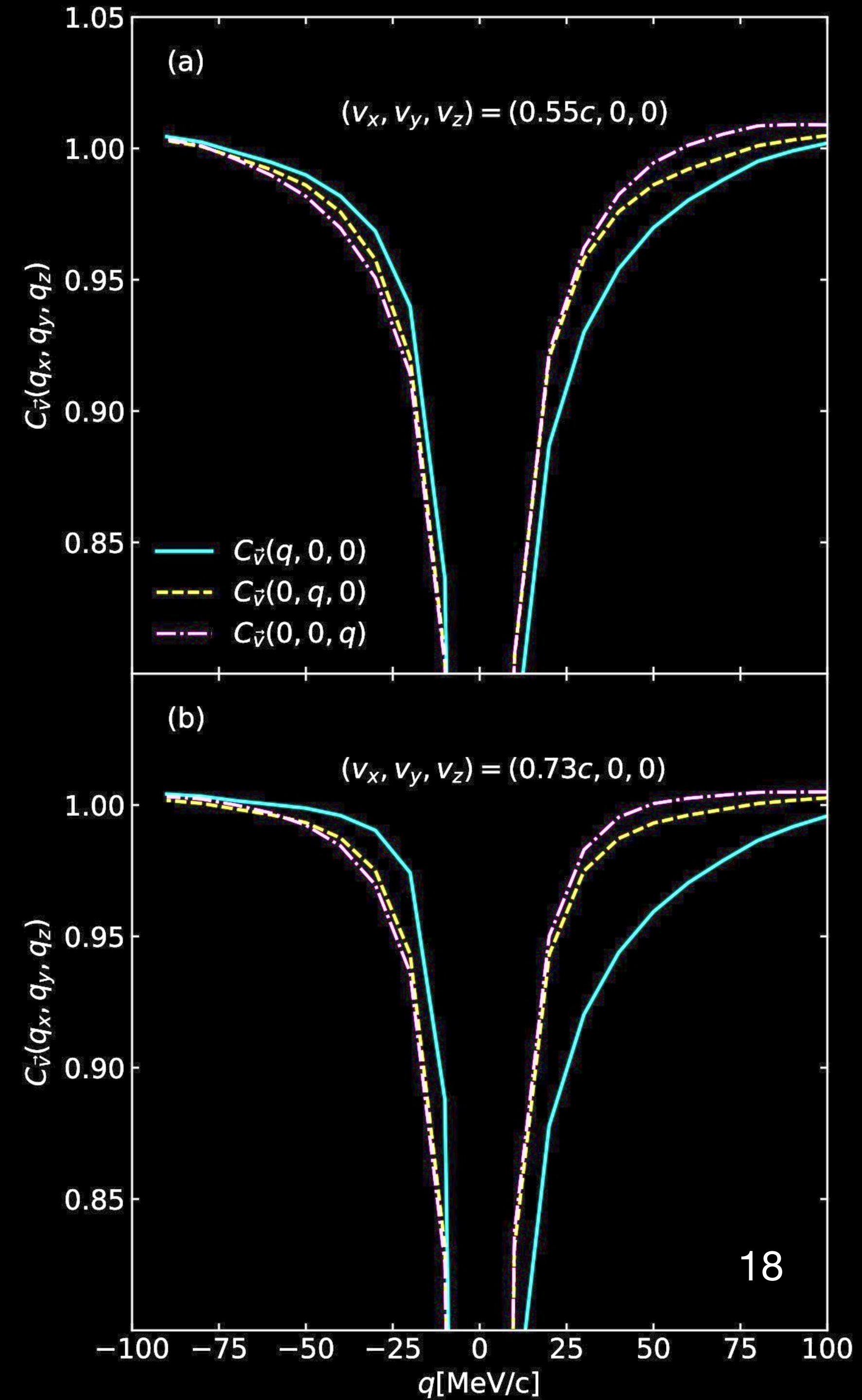
In experiments:

$$C_{\vec{v}}(\vec{q}) = \frac{\frac{d^6 P_{12}}{dp_1^3 dp_2^3}}{\frac{d^3 P_1}{dp_1^3} \frac{d^3 P_2}{dp_2^3}}, \quad \text{Ratio of two- to single-particle spectra}$$

Correlation functions

- Finite distance between protons and pions breaks inversion $\vec{r} \rightarrow -\vec{r}$ symmetry.
- This produces asymmetry in the correlation function $C(\vec{q}) \neq C(-\vec{q})$.
- Correlation functions can be measured in experiment and used in vorticity studies at low energies. At the energies where pions are scarce one can use proton duetron pairs.

$$C_{\vec{v}}(\vec{q}) = \frac{\frac{d^6 P_{12}}{dp_1^3 dp_2^3}}{\frac{d^3 P_1}{dp_1^3} \frac{d^3 P_2}{dp_2^3}}, \quad C_{\vec{v}}(\vec{q}) = \int d\vec{r} K(\vec{q}, \vec{r}) S_{\vec{v}}(\vec{r}).$$



Conclusions

- Vorticity of the fireball created in heavy-ion collisions was studied.
- The rotation modifies space-time picture of particle emission from the fireball.
- This leads to non-zero angle between ejection velocity and separation between different particle species, i.e. protons and pions, protons and neutrons.
- This changes momentum correlation functions directly measurable in experiments.
- This allows to conduct vorticity studies across virtually all energies, from GANIL and FRIB to RHIC and LHC.

[OS, P. Danielewicz, D. Kincses, A.Sorensen, arXiv:2510.10795]

[OS, arXiv:2510.22456]

